

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

Mid-Semester Examination
 Course No.: PHY 4241
 Course Title: Physics II

Summer Semester: 2021 - 2022
 Full Marks: 75
 Time: 90 Minutes

There are **03 (three)** questions. Answer **03 (three)** questions. The symbols have their usual meanings. Marks of each question and corresponding CO and PO are written in the brackets.

1. i. Find I in the circuit in Fig. 1(a).

8

(CO2)

(PO1, PO2)

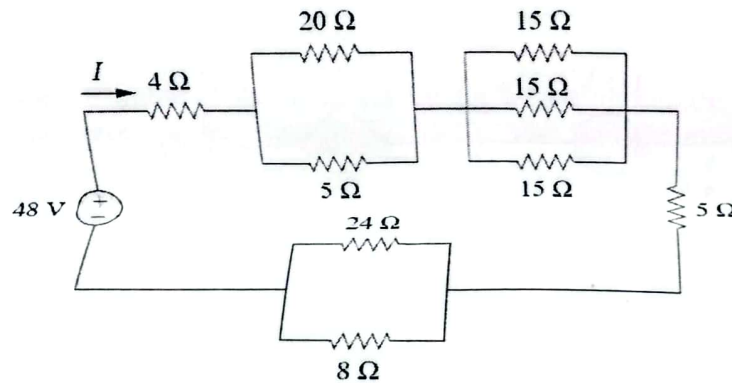


Fig. 1(a)

- ii. Find R_{eq} and i_o for the circuit in Fig. 1(b).

8

(CO2)

(PO1, PO2)

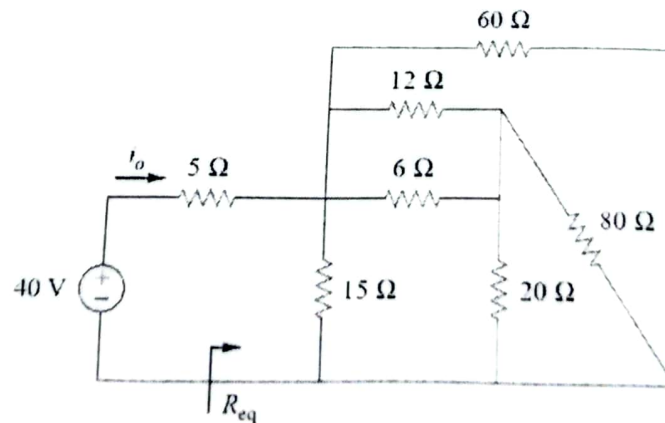


Fig. 1(b)

- iii. For the circuit in Fig. 1(c), find i_1 through i_4 .

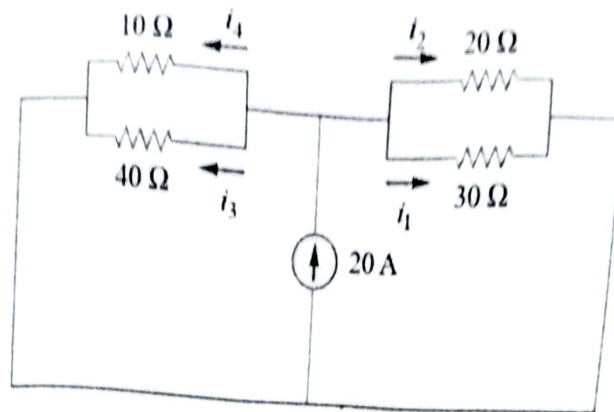


Fig. 1(c)

2. i. Find node voltages for the circuit in Fig. 2(a) using Nodal analysis.

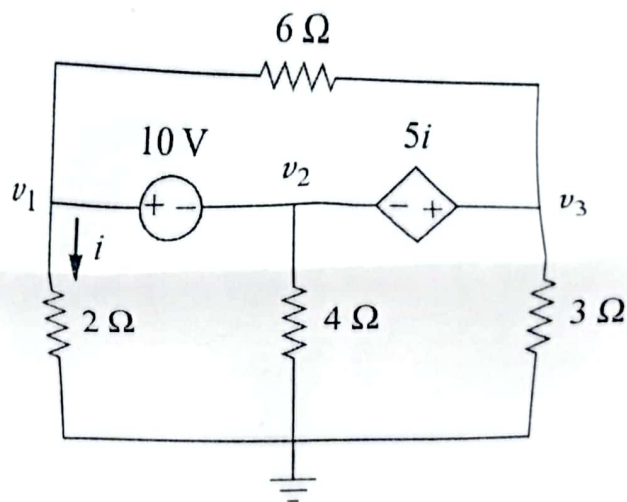


Fig. 2(a)

- ii. Determine i_0 for the circuit in Fig. 2(b) using Mesh analysis.

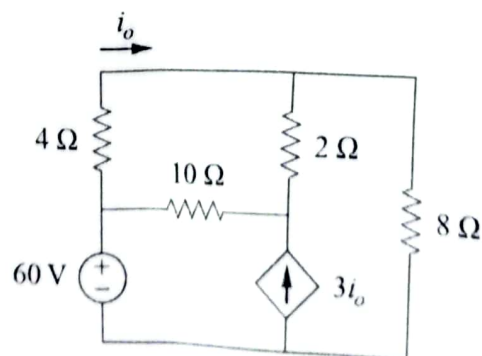


Fig. 2(b)

9

(CO2)
(PO1, PO2)

12

(CO2)
(PO1, PO2)

13

(CO2)
(PO1, PO2)

3. i. Find v_x for the circuit in Fig. 3(a) using source transformation. Show all the steps.

8
(CO2)
(PO1, PO2)

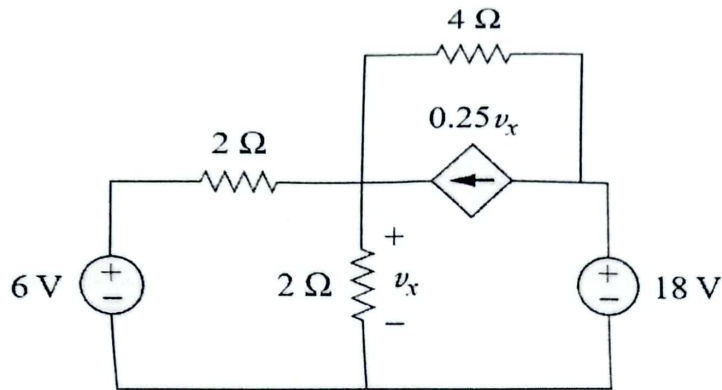


Fig. 3(a)

- ii. Determine the value of the load resistor for which maximum power will be delivered in the circuit in Fig. 3(b). Also find the maximum power.

8
(CO2)
(PO1, PO2)

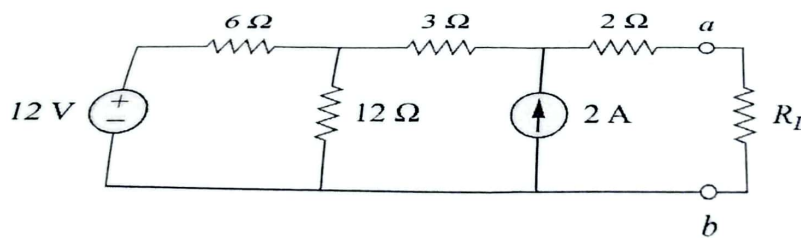


Fig. 3(b)

- iii. Find the Thevenin equivalent circuit of the circuit in Fig. 3(c).

9
(CO2)
(PO1, PO2)

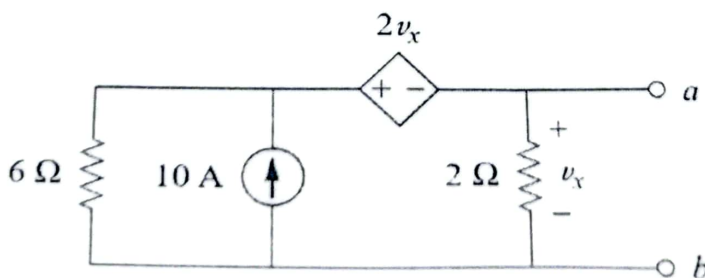


Fig. 3(c)